INTEGRATED MATERNAL AND CHILD HEALTH MONITORING SYSTEM

Final Capstone Design Document

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2.Problem Statement

Maternal and child healthcare services in many communities rely on paper-based cards, registers and separate record s that can be difficult to update, retrieve and share during routine care. As a result, mothers may miss antenatal appoin tments, important pregnancy measurements may not be reviewed promptly, children may miss scheduled immunizatio ns, and changes in a child's weight or height may not be noticed early. Lost or incomplete records can also make it diffi cult for mothers and health workers to confirm previous visits, vaccines received and follow-up actions required. Healt h workers may spend valuable time searching through registers and manually preparing overdue-patient lists and report s, reducing the time available for patient care. These problems can contribute to delayed follow-up and missed opportu nities to identify cases that need professional assessment. The proposed Integrated Maternal and Child Health Monito ring System is an online mobile application intended to bring essential maternal and child records into one organized di gital platform. It will allow authorized users to register mothers and children, record antenatal visits, immunizations, gro wth measurements and appointments, and review relevant histories. The system will support appointment and immuni zation reminders by SMS where an SMS gateway is available, and will identify records that meet configured follow-up r ules, such as overdue appointments or readings requiring review. Mothers will be able to view relevant appointments a nd child information, while health workers will manage clinical records and follow-up lists according to their roles. The system is intended to improve record accessibility, follow-up and continuity of care; it will support, not replace, professi onal clinical judgment and existing health-facility procedures.

Scope note: the case study is proposed as a selected health facility in WAKISO DISTRICT for example GODWIN DOCT OR'S HOSPITAL -NABBINGO, KABULENGWA NURSING HOME-KABULENGWA AND SINO DOCTORS CLINIC-BUKASA.

3. Case Study and Study Context

Proposed case study: maternal and child health service delivery at a selected health facilities in Wakiso. The study will examine how mothers' and children's records are registered, how antenatal visits and immunizations are documented, how growth measurements are followed over time, and how appointment reminders and overdue cases are managed.

3.3 Intended study methods

Subject to supervisor approval, information may be gathered through staff interviews, observation of the existing record workflow and review of blank/non-identifying forms and obtain required information.

4. Users and Their Goals

User	Goals
Mother / caregiver	View her maternal profile and appointments,see linked children; review vaccine dates and growth history,receive SMS reminders where available,know when to contact a health worker.
Health worker / nurse	Register mothers and children; record antenatal visits, immunizations and growth measurements,schedule appointments, identify overdue records and readings requiring clinical review.
Administrator	Create and deactivate user accounts, maintain authorized roles, review system activity and summary reports,support data quality, backups and configuration.

6. Functional Requirements (Detailed)

Functional requirements describe what the Integrated Maternal and Child Health Monitoring System must do.

- User Authentication and Login. The system shall allow authorized users to log in using a phone number and password, with email login supported if implemented. It shall validate user credentials and direct users to screens according to their roles.
- User Management. The system shall allow the administrator to create, edit, deactivate and reactivate user accounts and assign roles such as mother, health worker and administrator.
- Mother Registration. The system shall allow authorized health workers to register new mothers and update their profiles, including their names, contact details, dates of birth, addresses and expected delivery dates where applicable.
- Child Registration. The system shall allow authorized health workers to register children by entering their names, dates of birth and gender or sex, and linking each child to the correct mother's record.
- Antenatal Visit Recording. The system shall allow authorized health workers to record antenatal visits, including the visit date, gestational age where known, maternal weight, blood pressure and clinical notes.
- Immunization Recording. The system shall allow authorized health workers to record immunizations administered to children, including the vaccine name, dose number, date given and next due date where applicable.
- Child Growth Recording. The system shall allow authorized health workers to record children's growth measurements, including weight in kilograms, height or length in centimetres, and the date of measurement.
- Child Growth History. The system shall allow authorized users to view previous child growth measurements in chronological order and, where implemented, display the measurements in a growth chart.
- Appointment Management. The system shall allow authorized users to create and view appointments. Authorized staff shall be able to update appointment dates, times and statuses, including pending, attended, missed or cancelled.
- SMS Reminders. The system shall identify upcoming appointments and immunization due dates and prepare reminder messages for registered phone numbers. It shall send SMS reminders when a suitable SMS gateway or service is configured and available.
- Overdue Follow-up. The system shall display a list of overdue appointments and immunizations to authorized health workers, including the relevant dates and reasons for followup.
- Health Risk Flags. The system shall flag records that meet configured follow-up criteria, such as high blood pressure readings, to prompt review by qualified health workers. These flags shall not be treated as confirmed medical diagnoses.
- Record Search and History. The system shall allow authorized users to search for maternal and child records

using permitted identifiers and view the relevant medical histories.

- Reports Generation. The system shall generate basic reports for authorized administrators and health worker s, including the numbers of registered mothers and children, visits, immunizations and overdue follow-ups wit hin a selected period.
- Data Correction and Audit. The system shall validate required information and allow authorized users to corre ct errors. Where feasible, it shall record who created or modified a record and when the change occurs.
- Offline Data Capture and Synchronization. If offline functionality is implemented, the system shall allow user s to save records locally when the internet connection is unavailable and synchronize them with the server wh en connectivity returns, while avoiding duplicate reurred.

7. Non-Functional Requirements (Detailed)

Non-functional requirements describe how well the system should operate, including its security, usability, performanc e and reliability.

- Usability. The system shall provide simple and understandable interfaces, clear field labels, readable text and helpful validation messages so that users with basic smartphone skills can operate it effectively.
- Performance. The system shall aim to load common screens within approximately three seconds on a suppor ted device with a functioning local database. Network-dependent operations may take longer and shall display appropriate progress or error messages.
- Security. The system shall protect user accounts by storing passwords as secure hashes rather than plain tex t. Communications with a remote server shall use HTTPS, and access to records shall be restricted according to user roles.
- Privacy and Confidentiality. The system shall ensure that identifiable maternal and child health information is a ccessible only to authorized users. It shall avoid exposing sensitive medical information in unnecessary logs o r phone notifications and shall follow applicable data-protection requirements.
- Reliability and Data Integrity. The system shall validate required fields, dates and numerical measurements, m aintain correct relationships between database records, and clearly indicate whether saving or synchronization operations have succeeded or failed.
- Availability and Connectivity. The system shall tolerate slow or intermittent internet connections for features d esigned to work offline. SMS sending and remote synchronization shall depend on the availability of the neces sary network and services.
- Compatibility. The mobile application shall be developed and tested for the Android version agreed upon for the project and shall be tested on at least one supported physical Android device or emulator.
- Maintainability. The system shall use consistent database table names, field names, code organization and do cumentation so that future developers can understand, correct and extend the application more easily.
- Backup and Recovery. Where a server database is used, the system administrator shall arrange regular datab ase backups and test the restoration process to reduce the risk of permanent data loss.
- SMS Transparency. The system shall record whether SMS reminders are queued, sent or failed whenever the selected SMS provider supports delivery-status reporting. Users shall be informed about the purpose of collect ing their contact details for reminders.
- Scalability. The system shall be designed to support the addition of more users, mothers, children and health r ecords without requiring a complete redesign of the application. Additional facilities and facility-level reports may be incorporated as the system grows.
- Clinical Safety. The system shall use approved clinical guidance when configuring immunization schedules, g rowth interpretation and health-risk thresholds. It shall present risk flags as prompts for qualified health worke rs to review rather than as automatic diagnoses or treatment instructions.

8.1 Key use cases

- Mother: log in, view appointments, view child vaccine schedule and growth history, receive reminders.
- Health worker: register mother/child, record antenatal visit, record immunization, record growth, review overdue ca ses.
- Administrator: manage accounts and review summary reports.
- System/service: prepare reminders and identify configured follow-up conditions; external SMS delivery depends on a gateway.

9. Entity Relationship (ERD)

The ERD keeps the seven core tables requested for the current database. The recommended optional SMS_REMINDE RS table logs reminder messages and delivery status. The exact field names should be matched to the actual databas e before coding.

Proposed Entity Relationship Diagram (ERD)

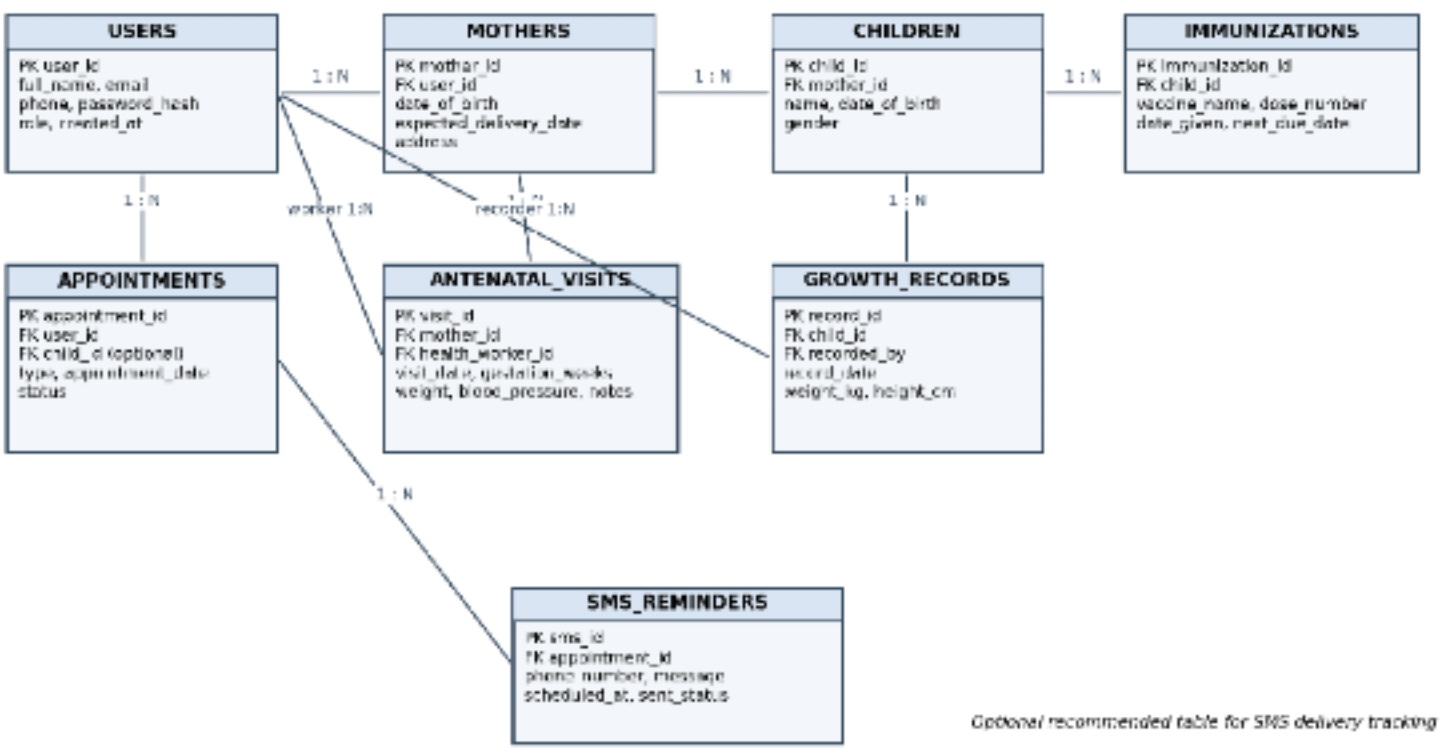


Figure 3. Proposed ERD for the core tables and optional SMS reminder log.

10. Data Dictionary

The following is a proposed data dictionary. INT primary keys should be auto-incrementing where supported. Foreign-k ey columns must use the same data type as the referenced primary key. Avoid storing real patient information in develo pment/test records.

11. Suggested screens for the first implementation

Screen	Main elements
Login / registration	Phone/email identifier, password, login button and registration rout e.
Mother home	Next appointment, children linked to the mother, upcoming reminde r and navigation.

Health-worker home	Search, visit/overdue counts and quick actions to register, record ANC, vaccination and growth.
Mother profile	Contact details, date of birth, address and expected delivery date where relevant.
Child profile	Name, date of birth, gender/sex and linked mother.
Antenatal visit	Mother selection, visit date, gestational age, weight, blood pressure, notes and save action.
Immunization	Child, vaccine/dose, date given, next due date and save action.
Growth record	Child, measurement date, weight, height/length and previous measurements.
Appointments and reminders	Appointment type/date/status and reminder status; SMS controls depend on gateway integration.
Admin / reports	User management and simple summary reports.

12. SMS Reminder Design

- The system should use the contact number saved in the authorized user's profile and should send only relevant appointment or immunization reminders.
- A reminder message should include the appointment date/time and facility information only where this is appropriate and agreed. Avoid unnecessary sensitive clinical details in SMS.
- A configured SMS gateway/provider, connectivity, valid recipient number and service credits may be required. The application should show a queued/sent/failed state where the provider supports it.
- The SMS_REMINDERS table is recommended if the project needs to track attempts. If the implementation remains small, reminder status may initially be stored against appointments, but this provides less complete history.
- Offline records can be queued locally, but SMS cannot normally be sent until the device/server has network access.

13. Validation, Security and Ethical Considerations

- Use sample or synthetic patient data for demonstrations unless the facility and supervisor have authorized use of real data.
- Restrict access by role and verify authorization on the server as well as hiding inappropriate app screens.
- Hash passwords; do not store plain-text passwords or put secrets/API keys directly into public source code.
- Validate dates, required fields, positive measurements and foreign-key selections before saving.
- Use appropriate clinical guidance for vaccine schedules and risk thresholds. The app should flag entries for qualified review, not make independent diagnoses.
- Confirm the case-study site, consent/permission process, privacy obligations and SMS messaging approach with the lecturer/supervisor.

14. Conclusion

The consolidated proposal is a mobile Integrated Maternal and Child Health Monitoring System that organizes maternal and child records and supports appointment follow-up. It retains the seven core database tables specified by the student, clarifies the main users and detailed requirements, and provides a unified case-study workflow, use-case diagram, ERD, data dictionary and screen sketches. SMS reminders and optional reminder tracking can improve follow-up when a suitable SMS service is available.